D3. Proposed Package Insert

Draft of the actual product proposed package insert is attached here.

MYTAFF 25 mg
TENOFOVIR ALAFENAMIDE TABLETS

NAME OF THE MEDICINAL PRODUCT

MYTAFF 25mg (Tenofovir Alafenamide Tablets)

QUALITATIVE AND QUANTITATIVE COMPOSITION

Each Film Coated Tablet contains:
Tenofovir Alafenamide fumarate equivalent to Tenofovir Alafenamide 25 mg

Excipients with known effects:
Each film-coated tablet contains 67.957 mg of lactose monohydrate.

PHARMACEUTICAL FORM

Film-coated tablet;
A white to off-white, film-coated, round, biconvex tablet debossed with M on one side of the tablet and TFI on other side.

CLINICAL PARTICULARS

Therapeutic indications

Tenofovir Alafenamide is indicated for the treatment of chronic hepatitis B in adults and adolescents (aged 12 years and older with body weight at least 35 kg) (see section Pharmacodynamic).

Posology and method of administration

Therapy should be initiated by a physician experienced in the management of chronic hepatitis B.

<u>Posology</u>
Adults and adolescents (aged 12 years and older with body weight at least 35 kg): one tablet once daily.

<u>Treatment discontinuation</u>
Treatment discontinuation may be considered as follows (See section Warning and Precautions):
•In HBeAg-positive patients without cirrhosis, treatment should be administered for at least 6-12 months after HBe seroconversion (HBeAg loss and HBV DNA loss with anti-HBe detection) is confirmed or until HBs seroconversion or until there is loss of efficacy. (See section Warning and Precautions). Regular reassessment is recommended after treatment discontinuation to detect virological relapse.
•In HBeAg-negative patients without cirrhosis, treatment should be administered at least until HBs seroconversion or until there is evidence of loss of efficacy. With prolonged treatment for more than 2 years, regular reassessment is recommended to confirm that continuing the selected therapy remains appropriate for the patient.

<u>Missed dose</u>
If a dose is missed and less than 18 hours have passed from the time it is usually taken, the patient should take Tenofovir alafenamide as soon as possible and then resume their normal dosing schedule. If more than 18 hours have passed from the time it is usually taken, the patient should not take the missed dose and should simply resume the normal dosing schedule. If the patient vomits within 1 hour of taking Tenofovir alafenamide , the patient should take another tablet. If the patient vomits more than 1 hour after taking Tenofovir alafenamide , the patient does not need to take another tablet.
<u>Special populations</u>

Elderly
No dose adjustment of Tenofovir alafenamide is required in patients aged 65 years and older (see section Pharmacokinetic)

Renal impairment
No dose adjustment of Tenofovir alafenamide is required in adults or adolescents (aged at least 12 years and of at least 35 kg body weight) with estimated creatinine clearance (CrCl) ≥ 15 mL/min or in patients with CrCl < 15 mL/min who are receiving haemodialysis.

On days of haemodialysis, Tenofovir alafenamide should be administered after completion of haemodialysis treatment (see section Pharmacokinetic)

No dosing recommendations can be given for patients with CrCl < 15 mL/min who are not receiving haemodialysis (See section Warning and Precautions)

Hepatic impairment
No dose adjustment of Tenofovir alafenamide is required in patients with hepatic impairment (See section Warning and Precautions & Pharmacokinetic)

Paediatric population
The safety and efficacy of Tenofovir alafenamide in children younger than 12 years of age, or weighing < 35 kg, have not yet been established. No data are available.

MODE OF ADMINISTRATION
Oral. Tenofovir alafenamide film-coated tablets should be taken with food

SIDE EFFECTS
Summary of the safety profile
The most frequently reported adverse reactions were headache, nausea, and fatigue.

Tabulated summary of adverse reactions
The following adverse drug reactions have been identified with tenofovir alafenamide in patients with chronic hepatitis B (Table 1). The adverse reactions are listed below by body system organ class and frequency. Frequencies are defined as follows: very common, common, uncommon, rare or very rare.

Table 1: Adverse drug reactions identified with tenofovir alafenamide

System organ class	
Frequency	**Adverse reaction**
Gastrointestinal disorders	
Common	Diarrhoea, vomiting, nausea, abdominal pain, abdominal distension, flatulence
General disorders and administration site conditions	
Common	Fatigue
Nervous system disorders	
Very common	Headache
Common	Dizziness
Skin and subcutaneous tissue disorders	

Common	Rash, pruritus
Hepatobiliary disorders	
Common	Increased ALT
Musculoskeletal and connective tissue disorders	
Common	Arthralgia

DRUG INTERACTIONS
Interaction studies have only been performed in adults. Tenofovir alafenamide should not be co-administered with medicinal products containing tenofovir disoproxil fumarate, tenofovir alafenamide or adefovir dipivoxil.

<u>Medicinal products that may affect tenofovir alafenamide</u>
Tenofovir alafenamide is transported by P-gp and breast cancer resistance protein (BCRP). Medicinal products that are P-gp inducers (e.g., rifampicin, rifabutin, carbamazepine, phenobarbital or St. John's wort) are expected to decrease plasma concentrations of tenofovir alafenamide, which may lead to loss of therapeutic effect of Tenofovir alafenamide . Co-administration of such medicinal products with Tenofovir alafenamide is not recommended. Co-administration of Tenofovir alafenamide with medicinal products that inhibit P-gp and/or BCRP may increase plasma concentration of tenofovir alafenamide. Co-administration of strong inhibitors of P-gp with Tenofovir alafenamide is not recommended. Tenofovir alafenamide is a substrate of OATP1B1 and OATP1B3. The distribution of tenofovir alafenamide in the body may be affected by the activity of OATP1B1 and/or OATP1B3.

<u>Effect of tenofovir alafenamide on other medicinal products</u>
Tenofovir alafenamide is not an inhibitor of CYP1A2, CYP2B6, CYP2C8, CYP2C9, CYP2C19, or CYP2D6. It is not an inhibitor of CYP3A. Tenofovir alafenamide is not an inhibitor of human uridine diphosphate glucuronosyltransferase (UGT) 1A1. It is not known whether tenofovir alafenamide is an inhibitor of other UGT enzymes.

Drug interaction information for Tenofovir alafenamide with potential concomitant medicinal products is summarised in Table 2 below (increase is indicated as "↑", decrease as "↓", no change as "↔"; twice daily as "b.i.d.", single dose as "s.d.", once daily as "q.d."; and intravenously as "IV").

Table 2: Interactions between Tenofovir alafenamide and other medicinal products

Medicinal product by therapeutic areas	**Effects on drug levels.**	**Recommendation concerning co-administration with Tenofovir alafenamide**
	ANTICONVULSANTS	
Carbamazepine (300 mg orally, b.i.d.) Tenofovir alafenamide[a] (25 mg orally, s.d.)	*Tenofovir alafenamide* ↓ Cmax ↓ AUC *Tenofovir* ↓ Cmax ↔ AUC	Co-administration is not recommended.
Oxcarbazepine Phenobarbital	Interaction not studied. *Expected:* ↓ Tenofovir alafenamide	Co-administration is not recommended.

Phenytoin	Interaction not studied. *Expected:* ↓ Tenofovir alafenamide	Co-administration is not recommended.
Midazolam[b] (2.5 mg orally, s.d.) Tenofovir alafenamide[a] (25 mg orally, q.d.)	*Midazolam* ↔ Cmax ↔ AUC	No dose adjustment of midazolam (administered orally or IV) is required.
Midazolam[b] (1 mg IV, s.d.) Tenofovir alafenamide[a] (25 mg orally, q.d.)	*Midazolam* ↔ Cmax ↔ AUC	
ANTIDEPRESSANTS		
Sertraline (50 mg orally, s.d.) Tenofovir alafenamide[c] (10 mg orally, q.d.)	*Tenofovir alafenamide* ↔ Cmax ↔ AUC *Tenofovir* ↔ Cmax ↔ AUC ↔ Cmin	No dose adjustment of Tenofovir alafenamide or sertraline is required.
Sertraline (50 mg orally, s.d.) Tenofovir alafenamide[c] (10 mg orally, q.d.)	*Sertraline* ↔ Cmax ↔ AUC	
ANTIFUNGALS		
Itraconazole Ketoconazole	Interaction not studied. *Expected:* ↑ Tenofovir alafenamide	Co-administration is not recommended.
ANTIMYCOBACTERIALS		
Rifampicin Rifapentine	Interaction not studied. *Expected:* ↓ Tenofovir alafenamide	Co-administration is not recommended.
Rifabutin	Interaction not studied. *Expected:* ↓ Tenofovir alafenamide	Co-administration is not recommended.
HCV ANTIVIRAL AGENTS		
Sofosbuvir (400 mg orally, q.d.)	Interaction not studied. *Expected:* ↔Sofosbuvir ↔ GS-331007	No dose adjustment of Tenofovir alafenamide or sofosbuvir is required.
Ledipasvir/sofosbuvir (90 mg/400 mg orally, q.d.) Tenofovir alafenamide[d] (25 mg orally, q.d.)	*Ledipasvir* ↔ Cmax ↔ AUC ↔ Cmin *Sofosbuvir*	No dose adjustment of Tenofovir alafenamide or ledipasvir/sofosbuvir is required.

	↔ Cmax ↔ AUC *GS-331007* [e] ↔ Cmax ↔ AUC ↔ Cmin *Tenofovir alafenamide* ↔ Cmax ↔ AUC *Tenofovir* ↑ Cmax ↑ AUC ↑ Cmin	
Sofosbuvir/velpatasvir (400 mg/100 mg orally, q.d.)	Interaction not studied. *Expected:* — Sofosbuvir ↔ GS-331007 — Velpatasvir ↑ Tenofovir alafenamide	No dose adjustment of Tenofovir alafenamide or sofosbuvir/velpatasvir is required.
Sofosbuvir/velpatasvir/ voxilaprevir (400 mg/100 mg/ 100 mg + 100 mg[g] orally, q.d.) Tenofovir alafenamide[d] (25 mg orally, q.d.)	*Sofosbuvir* ↔ Cmax ↔ AUC *GS-331007*[e] ↔ Cmax ↔ AUC *Velpatasvir* ↔ Cmax ↔ AUC ↔ Cmin *Voxilaprevir* ↔ Cmax ↔ AUC ↔ Cmin *Tenofovir alafenamide* ↑ Cmax ↑ AUC	No dose adjustment of Tenofovir alafenamide or sofosbuvir/velpatasvir/voxilaprevir is required.
HIV ANTIRETROVIRAL AGENTS – PROTEASE INHIBITORS		
Atazanavir/cobicistat (300 mg/150 mg orally, q.d.) Tenofovir alafenamide[a] (10 mg orally, q.d.)	*Tenofovir alafenamide* ↑ Cmax ↑ AUC *Tenofovir* ↑ Cmax ↑ AUC ↑ Cmin *Atazanavir* ↔ Cmax ↔ AUC ↔ Cmin *Cobicistat* ↔ Cmax ↔ AUC ↑ Cmin	Co-administration is not recommended.

Atazanavir/ritonavir (300 mg/100 mg orally, q.d.) Tenofovir alafenamide[a] (10 mg orally, s.d.)	*Tenofovir alafenamide* ↑ Cmax ↑ AUC *Tenofovir* ↑ Cmax ↑ AUC *Atazanavir* ↔ Cmax ↔ AUC ↔ Cmin	Co-administration is not recommended.
Darunavir/cobicistat (800 mg/150 mg orally, q.d.) Tenofovir alafenamide[a] (25 mg orally, q.d.)	*Tenofovir alafenamide* ↔ Cmax ↔ AUC *Tenofovir* ↑ Cmax ↑ AUC ↑ Cmin *Darunavir* ↔ Cmax ↔ AUC ↔ Cmin *Cobicistat* ↔ Cmax ↔ AUC ↔ Cmin	Co-administration is not recommended.
Darunavir/ritonavir (800 mg/100 mg orally, q.d.) Tenofovir alafenamide[a] (10 mg orally, s.d.)	*Tenofovir alafenamide* ↑ Cmax ↔ AUC *Tenofovir* ↑ Cmax ↑ AUC *Darunavir* ↔ Cmax ↔ AUC ↔ Cmin	Co-administration is not recommended.
Lopinavir/ritonavir (800 mg/200 mg orally, q.d.) Tenofovir alafenamide[a] (10 mg orally, s.d.)	*Tenofovir alafenamide* ↑ Cmax ↑ AUC *Tenofovir* ↑ Cmax ↑ AUC *Lopinavir* ↔ Cmax ↔ AUC ↔ Cmin	Co-administration is not recommended.
Tipranavir/ritonavir	Interaction not studied. *Expected:* ↓ Tenofovir alafenamide	Co-administration is not recommended.
HIV ANTIRETROVIRAL AGENTS – INTEGRASE INHIBITORS		

Dolutegravir (50 mg orally, q.d.) Tenofovir alafenamide[a] (10 mg orally, s.d.)	*Tenofovir alafenamide* ↑ Cmax ↑ AUC *Tenofovir* ↔ Cmax ↑ AUC *Dolutegravir* ↔ Cmax ↔ AUC ↔ Cmin	No dose adjustment of Tenofovir alafenamide or dolutegravir is required.
Raltegravir	Interaction not studied. *Expected:* ↔Tenofovir alafenamide ↔Raltegravir	No dose adjustment of Tenofovir alafenamide or raltegravir is required.
HIV ANTIRETROVIRAL AGENTS – NON-NUCLEOSIDE REVERSE TRANSCRIPTASE INHIBITORS		
Efavirenz (600 mg orally, q.d.) Tenofovir alafenamide[f] (40 mg orally, q.d.)	*Tenofovir alafenamide* ↓ Cmax ↔ AUC *Tenofovir* ↓ Cmax ↔ AUC ↔ Cmin *Expected:* ↔Efavirenz	No dose adjustment of Tenofovir alafenamide or efavirenz is required.
Nevirapine	Interaction not studied. *Expected:* ↔Tenofovir alafenamide ↔Nevirapine	No dose adjustment of Tenofovir alafenamide or nevirapine is required.
Rilpivirine (25 mg orally, q.d.) Tenofovir alafenamide (25 mg orally, q.d.)	*Tenofovir alafenamide* ↔ Cmax ↔ AUC *Tenofovir* ↔ Cmax ↔ AUC ↔ Cmin *Rilpivirine* ↔ Cmax ↔ AUC ↔ Cmin	No dose adjustment of Tenofovir alafenamide or rilpivirine is required.
HIV ANTIRETROVIRAL AGENTS – CCR5 RECEPTOR ANTAGONIST		
Maraviroc	Interaction not studied. *Expected:* ↔Tenofovir alafenamide ↔Maraviroc	No dose adjustment of Tenofovir alafenamide or maraviroc is required.
HERBAL SUPPLEMENTS		

St. John's wort *(hypericum perforatum)*	Interaction not studied. *Expected:* ↓ Tenofovir alafenamide	Co-administration is not recommended.
ORAL CONTRACEPTIVES		
Norgestimate (0.180 mg/0.215 mg/ 0.250 mg orally, q.d.) Ethinyl estradiol (0.025 mg orally, q.d.) Tenofovir alafenamide[a] (25 mg orally, q.d.)	*Norelgestromin* ↔ Cmax ↔ AUC ↔ Cmin *Norgestrel* ↔ Cmax ↔ AUC ↔ Cmin *Ethinyl estradiol* ↔ Cmax ↔ AUC ↔ Cmin	No dose adjustment of Tenofovir alafenamide or norgestimate/ethinyl estradiol is required.

a. Used as Emtricitabine/tenofovir alafenamide fixed-dose combination tablet
b. A sensitive CYP3A4 substrate
c. Used as Elvitegravir/cobicistat/emtricitabine/tenofovir alafenamide fixed-dose combination tablet
d. Used as Emtricitabine/rilpivirine/tenofovir alafenamide fixed-dose combination tablet
e. The predominant circulating nucleoside metabolite of sofosbuvir
f. Used as Tenofovir alafenamide 40 mg and emtricitabine 200 mg
g. Used with additional voxilaprevir 100 mg to achieve voxilaprevir exposures expected in HCV infected patients.

WARNINGS AND PRECAUTIONS

HBV transmission

Patients must be advised that Tenofovir alafenamide does not prevent the risk of transmission of HBV to others through sexual contact or contamination with blood. Appropriate precautions must continue to be used.

Patients with decompensated liver disease

There are no data on the safety and efficacy of Tenofovir alafenamide in HBV-infected patients with decompensated liver disease and who have a Child Pugh Turcotte (CPT) score > 9 (i.e. class C). These patients may be at higher risk of experiencing serious hepatic or renal adverse reactions. Therefore, hepatobiliary and renal parameters should be closely monitored in this patient population

Exacerbation of hepatitis

Flares on treatment

Spontaneous exacerbations in chronic hepatitis B are relatively common and are characterised by transient increases in serum alanine aminotransferase (ALT). After initiating antiviral therapy, serum ALT may increase in some patients. In patients with compensated liver disease, these increases in serum ALT are generally not accompanied by an increase in serum bilirubin concentrations or hepatic decompensation. Patients with cirrhosis may be at a higher risk for hepatic decompensation following hepatitis exacerbation, and therefore should be monitored closely during therapy.

Flares after treatment discontinuation

Acute exacerbation of hepatitis has been reported in patients who have discontinued treatment for hepatitis B, usually in association with rising HBV DNA levels in plasma. The majority of cases are self-limited but severe exacerbations, including fatal outcomes, may occur after discontinuation of treatment for hepatitis B. Hepatic function should be monitored at repeated intervals with both clinical and laboratory follow-up for at least 6 months after discontinuation of treatment for hepatitis B. If appropriate, resumption of hepatitis B therapy may be warranted. In patients with advanced liver disease or cirrhosis, treatment discontinuation is not recommended since post-treatment exacerbation of hepatitis may lead to hepatic decompensation. Liver flares are especially serious, and sometimes fatal in patients with decompensated liver disease.

Renal impairment

Patients with creatinine clearance < 30 mL/min

The use of Tenofovir alafenamide once daily in patients with CrCl ≥ 15 mL/min but < 30 mL/min and in patients with CrCl < 15 mL/min who are receiving haemodialysis is based on very limited pharmacokinetic data and on modelling and simulation. There are no safety data on the use of Tenofovir alafenamide to treat HBV-infected patients with CrCl < 30 mL/min.

The use of Tenofovir Alafenamide is not recommended in patients with CrCl < 15 mL/min who are not receiving haemodialysis

Nephrotoxicity

A potential risk of nephrotoxicity resulting from chronic exposure to low levels of tenofovir due to dosing with tenofovir alafenamide cannot be excluded

Patients co-infected with HBV and hepatitis C or D virus

There are no data on the safety and efficacy of Tenofovir alafenamide in patients co-infected with hepatitis C or D virus. Co-administration guidance for the treatment of hepatitis C should be followed

Hepatitis B and HIV co-infection

HIV antibody testing should be offered to all HBV-infected patients whose HIV-1 infection status is unknown before initiating therapy with Tenofovir alafenamide. In patients who are co-infected with HBV and HIV, Tenofovir alafenamide should be co-administered with other antiretroviral agents to ensure that the patient receives an appropriate regimen for treatment of HIV

Co-administration with other medicinal products

Tenofovir alafenamide should not be co-administered with products containing tenofovir alafenamide, tenofovir disoproxil fumarate or adefovir dipivoxil.
Co-administration of Tenofovir alafenamide with certain anticonvulsants (e.g. carbamazepine, oxcarbazepine, phenobarbital and phenytoin), antimycobacterials (e.g. rifampicin, rifabutin and rifapentine) or St. John’s wort, all of which are inducers of P-glycoprotein (P-gp) and may decrease tenofovir alafenamide plasma concentrations, is not recommended.
Co-administration of Tenofovir alafenamide with strong inhibitors of P-gp (e.g. itraconazole and ketoconazole) may increase tenofovir alafenamide plasma concentrations. Co-administration is not recommended.

Lactose intolerance

MYTAFF contains lactose monohydrate. Consequently, patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency, or glucose-galactose malabsorption should not take this medicinal product.

OVERDOSE

If overdose occurs the patient must be monitored for evidence of toxicity. Treatment of overdose with Tenofovir alafenamide consists of general supportive measures including monitoring of vital signs as well as observation of the clinical status of the patient. Tenofovir is efficiently removed by haemodialysis with an extraction coefficient of approximately 54%. It is not known whether tenofovir can be removed by peritoneal dialysis.

CONTRAINDICATIONS

Hypersensitivity to the active substance or to any of the excipients

PREGNANCY AND LACTATION

Pregnancy

There are no or limited amount of data (less than 300 pregnancy outcomes) from the use of tenofovir alafenamide in pregnant women. However, a large amount of data on pregnant women (more than 1000 exposed outcomes) indicates no malformative nor feto/neonatal toxicity associated with the use of tenofovir disoproxil fumarate. Animal studies do not indicate direct or indirect harmful effects with respect to reproductive toxicity. The use of Tenofovir alafenamide may be considered during pregnancy, if necessary.

Breast-feeding

It is not known whether tenofovir alafenamide is secreted in human milk. However, in animal studies it has been shown that tenofovir is secreted into milk. There is insufficient information on the effects of tenofovir in newborns/infants. A risk to the breastfed child cannot be excluded; therefore, Tenofovir alafenamide should not be used during breast-feeding.

Fertility

No human data on the effect of Tenofovir alafenamide on fertility are available. Animal studies do not indicate harmful effects of tenofovir alafenamide on fertility.

PHARMACOLOGY

Pharmacotherapeutic group: Antiviral for systemic use, nucleoside and nucleotide reverse transcriptase inhibitors; ATC code: J05AF13.

Mechanism of action

Tenofovir alafenamide is a phosphonamidate prodrug of tenofovir (2'-deoxyadenosine monophosphate analogue). Tenofovir alafenamide enters primary hepatocytes by passive diffusion and by the hepatic uptake transporters OATP1B1 and OATP1B3. Tenofovir alafenamide is primarily hydrolyzed to form tenofovir by carboxylesterase 1 in primary hepatocytes. Intracellular tenofovir is subsequently phosphorylated to the pharmacologically active metabolite tenofovir diphosphate. Tenofovir diphosphate inhibits HBV replication through incorporation into viral DNA by the HBV reverse transcriptase, which results in DNA chain termination.

Tenofovir has activity that is specific to hepatitis B virus and human immunodeficiency virus (HIV-1 and HIV-2). Tenofovir diphosphate is a weak inhibitor of mammalian DNA polymerases that include mitochondrial DNA polymerase γ and there is no evidence of mitochondrial toxicity in vitro based on several assays including mitochondrial DNA analyses.

Antiviral activity

The antiviral activity of tenofovir alafenamide was assessed in HepG2 cells against a panel of HBV clinical isolates representing genotypes A-H. The EC50 (50% effective concentration) values for tenofovir alafenamide ranged from 34.7 to 134.4 nM, with an overall mean EC50 of 86.6 nM. The CC50 (50% cytotoxicity concentration) in HepG2 cells was > 44400 nM.

Resistance

No amino acid substitutions associated with resistance to Tenofovir alafenamide were identified in these isolates(genotypic and phenotypic analyses) .

Cross-resistance

The antiviral activity of tenofovir alafenamide was evaluated against a panel of isolates containing nucleos(t)ide reverse transcriptase inhibitor mutations in HepG2 cells. HBV isolates expressing the rtV173L, rtL180M, and rtM204V/I substitutions associated with resistance to lamivudine remained susceptible to tenofovir alafenamide (< 2-fold change in EC50). HBV isolates expressing the rtL180M, rtM204V plus rtT184G, rtS202G, or rtM250V substitutions associated with resistance to entecavir remained susceptible to tenofovir alafenamide. HBV isolates expressing the rtA181T, rtA181V, or rtN236T single substitutions associated with resistance to adefovir remained susceptible to tenofovir alafenamide; however, the HBV isolate expressing rtA181V plus rtN236T exhibited reduced susceptibility to tenofovir alafenamide (3.7-fold change in EC50). The clinical relevance of these substitutions is not known.

PHARMACOKINETICS

Absorption

Following oral administration of Tenofovir alafenamide under fasted conditions in adult patients with chronic hepatitis B, peak plasma concentrations of tenofovir alafenamide were observed approximately 0.48 hours post-dose. In patients with CHB, mean steady state AUC0-24 for tenofovir alafenamide and tenofovir were 0.22 µg•hr/mL and 0.32 µg•hr/mL, respectively. Steady state Cmax for tenofovir alafenamide and tenofovir were 0.18 and 0.02 µg/mL, respectively. Relative to fasting conditions, the administration of a single dose of Tenofovir alafenamide with a high fat meal resulted in a 65% increase in tenofovir alafenamide exposure.

Distribution

The binding of tenofovir alafenamide to human plasma proteins was approximately 80%. The binding of tenofovir to human plasma proteins is less than 0.7% and is independent of concentration over the range of 0.01–25 µg/mL.

Biotransformation

Metabolism is a major elimination pathway for tenofovir alafenamide in humans, accounting for > 80% of an oral dose. It is shown that tenofovir alafenamide is metabolized to tenofovir (major metabolite) by carboxylesterase-1 in hepatocytes; and by cathepsin A in peripheral blood mononuclear cells (PBMCs) and macrophages. Tenofovir alafenamide is hydrolysed within cells to form tenofovir (major metabolite), which is phosphorylated to the active metabolite, tenofovir diphosphate. Tenofovir alafenamide is not metabolized by CYP1A2, CYP2C8, CYP2C9, CYP2C19, or CYP2D6. Tenofovir alafenamide is minimally metabolized by CYP3A4.

Elimination

Renal excretion of intact tenofovir alafenamide is a minor pathway with < 1% of the dose eliminated in urine. Tenofovir alafenamide is mainly eliminated following metabolism to tenofovir. Tenofovir alafenamide and tenofovir have a median plasma half-life of 0.51 and 32.37 hours, respectively. Tenofovir is renally eliminated from the body by the kidneys by both glomerular filtration and active tubular secretion.

Linearity/non-linearity

Tenofovir alafenamide exposures are dose proportional over the dose range of 8 to 125 mg.

Pharmacokinetics in special populations

Age, gender and ethnicity

No clinically relevant differences in pharmacokinetics according to age or ethnicity have been identified. Differences in pharmacokinetics according to gender were not considered to be clinically relevant.

Hepatic impairment

In patients with severe hepatic impairment, total plasma concentrations of tenofovir alafenamide and tenofovir are lower than those with normal hepatic function. When corrected for protein binding, unbound (free) plasma concentrations of tenofovir alafenamide in severe hepatic impairment and normal hepatic function are similar.

Renal impairment

No clinically relevant differences in tenofovir alafenamide or tenofovir pharmacokinetics were observed between healthy subjects and patients with severe renal impairment (estimated CrCl > 15 but < 30 mL/min)

Paediatric population

No clinically relevant differences in tenofovir alafenamide or tenofovir pharmacokinetics were observed between adolescent and adult HIV-1-infected subjects.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINE

Tenofovir alafenamide has no or negligible influence on the ability to drive and use machines. Patients should be informed that dizziness has been reported during treatment with Tenofovir alafenamide

PARTICULARS

List of Excipients

Core tablet

Lactose Monohydrate, Microcrystalline Cellulose, Croscarmellose Sodium, Magnesium Stearate

Film coat

Polyvinyl Alcohol, Polyethylene Glycol, Titanium Dioxide & Talc.

Incompatibilities

Not applicable.

Shelf life

36 months

Special precautions for storage

Store in a dry place below 30°C. Store in the original container

Nature and contents of container

HDPE Bottle of 30’s pack

Special precautions for disposal

No special requirements. Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

PRODUCT REGISTRATION HOLDER

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